

Summer Training Program WEEK 2 - DAY 5

- Operators
- Conditional constructs

Operators: Used to perform computation.

Arithmetic operators

The following are arithmetic operators in python:

+	Addition
-	Subtraction
*	Multiply
/	division
**	raised to the power/exponential
//	integral division(only returns the whole part of the division result)

Assignment operators

=
+=
-=
*=
/=

**=
//=

- All the assignment in python is dynamic (dynamic memory allocation)
- Python does not have any increment or decrement operators

Conditional operators

They return either true or false value

>	greater than	
<	less than	
>=	greater than equal to	EDGE CASE
<=	less than equal to	EDGE CASE
==	is equal to (check)	
!=	not equal to	

Membership operators

Also returns true or false value

is
in
not in
is not

Boolean operators

True

False

These are mainly used in toggles, where conditions are mainly used.

For eg:

When gps system is to be used:

Both internet and gps system must be enabled

So such loops are used:

if internet true:

code

If gps true:

code

Class examples:

Activity status

File permissions

Face recognition

Logical operators

and
or
not

Bitwise operators

AND
OR
NOT
XOR
LEFT SHIFT
RIGHT SHIFT

The main purpose of bitwise operators is that they are used for the purpose of data encryption.

Finding inverse of a matrix, $a \cdot a^{-1} = i$ is an encryption/decryption technique

Conditional constructs

if	the decision is taken at one point, start of the construct
if elif else	multiple broken down decision scenarios
nested if else	

Interview questions:

- What are bitwise operators and where are they used?
- Static vs dynamic memory allocation

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ASSIGNMENT:

- **Explore PEP (python enhancement proposal)**
- **Explore AES and SHA-256 security algorithms**
- **Write a Python program to simulate an online shopping coupon system.**
- **Code for the problem of one brick in a wall**

(PUSHED TO REPO)

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